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DUAL PATH CLOCK FORWARDING

Abstract

Retimers and retiming methods may employ dual path clock forwarding to drive a transmit clock generator. One illustrative integrate retimer circuit includes: a sampling element configured to produce a digital receive signal by sampling an analog receive signal in accordance with a sampling signal; a timing error estimator configured to produce a timing error signal indicating an estimated timing error of the sampling signal relative to the analog receive signal; a clock recovery circuit configured to derive the sampling signal from the estimated timing error and a reference clock in part by determining a frequency signal; a transmitter configured to retransmit the digital receive signal in accordance with a transmit clock; and a transmit clock generator configured to derive the transmit clock. The transmit clock generator operates based on each of: the reference clock; the frequency signal; and a phase error of the transmit clock relative to the sampling signal.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to digital retimers, and more particularly, to retimers using clock recovery circuits and techniques to drive a transmit clock generator with dual path clock forwarding.

BACKGROUND

[0002] As high-speed data streams travel from source to destination, they are often received and re-transmitted to, e.g., combat attenuation, transition between different links, or perhaps traverse intermediate devices. Digital retimers perform such receiving and retransmission using clock recovery, symbol detection, and buffering. Because the buffer memory is limited, the retransmission clock is tied directly or indirectly (e.g., via the buffer fill level) to the recovered receive clock. As a consequence, jitter in the receive clock often causes jitter in the retransmission clock. A careful analysis of the jitter transfer function gain in existing designs reveals peaks that undesirably impair system performance, particularly when combined with transfer function peaking of the clock recovery circuit and/or retransmission clock phase lock loop that may result from process, voltage, and temperature (PVT) variations.

SUMMARY

[0003] Accordingly, there are disclosed herein retimers and retiming methods employing dual path clock forwarding to drive a transmit clock generator. One illustrative integrate retimer circuit includes: a sampling element configured to produce a digital receive signal by sampling an analog receive signal in accordance with a sampling signal; a timing error estimator configured to produce a timing error signal indicating an estimated timing error of the sampling signal relative to the analog receive signal; a clock recovery circuit configured to derive the sampling signal from the estimated timing error and a reference clock in part by determining a frequency signal; a transmitter configured to retransmit the digital receive signal in accordance with a transmit clock; and a transmit clock generator configured to derive the transmit clock. The transmit clock generator operates based on each of: the reference clock; the frequency signal; and a phase error of the transmit clock relative to the sampling signal.

[0004] An illustrative retiming method includes: sampling an analog receive signal in accordance with a sampling signal to obtain a digital receive signal; producing a timing error signal indicating an estimated timing error of the sampling signal relative to the analog receive signal; deriving the sampling signal from the estimated timing error and a reference clock in part by determining a frequency signal; retransmitting the digital receive signal in accordance with a transmit clock; and deriving the transmit clock based on each of: the reference clock, the frequency signal, and a phase error of the transmit clock relative to the sampling signal.

[0005] An illustrative semiconductor intellectual property core generates circuitry for implementing a retimer and/or retiming method as described above.

[0006] Each of the foregoing retimer, retiming method, and core implementations may be embodied individually or conjointly and may be combined with any one or more of the following optional features: 1. the transmit clock generator includes a clock forwarding filter having: a phase comparator to determine the phase error; a phase filter coupled to the phase comparator to derive a filtered phase error; and a frequency filter to derive a filtered frequency signal. 2. the frequency signal is a division ratio for a fractional-N phase lock loop in a clock signal path for deriving the sampling signal from the reference clock. 3. the transmit clock generator includes a fractional-N phase lock loop in a clock signal path for deriving the transmit clock from the reference clock. 4. the frequency signal is a frequency offset for a phase interpolator in a clock signal path for deriving the sampling signal from the reference clock. 5. the transmit clock generator includes a phase interpolator in a clock signal path for deriving the transmit clock from the reference clock. 6. the phase filter includes a jitter filter. 7. the phase filter is a second-order filter. 8. the phase error is a difference between a write pointer for a buffer and a read pointer for the buffer. 9. a secondary clock forwarding filter that aligns a secondary transmit clock to said transmit clock.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** shows an illustrative network.

[0008] FIG. **2** is a block diagram of an illustrative switch.

[0009] FIGS. **3**A-**3**C are block diagrams of illustrative digital communications receivers having various clock recovery implementations.

[0010] FIG. **4** is a block diagram of an illustrative digital retimer with clock forwarding.

[0011] FIG. **5** is a block diagram of an illustrative digital retimer with dual path clock forwarding.

[0012] FIG. **6** is a block diagram of a first illustrative dual path forwarding filter implementation.

[0013] FIG. **7** shows an alternative phase comparator implementation.

[0014] FIG. **8** is a block diagram of a second illustrative dual path clock forwarding filter implementation.

[0015] FIG. **9** is a block diagram of a third illustrative dual path clock forwarding filter implementation.

[0016] FIG. **10** shows an alternative implementation of the transmit phase lock loop.

[0017] FIG. **11** shows the first illustrative dual path forwarding filter in a multi-channel retimer.

[0018] FIG. **12** shows the second illustrative dual path forwarding filter in a multi-channel retimer.

[0019] FIG. **13** shows the third illustrative dual path forwarding filter in a multi-channel retimer.

DETAILED DESCRIPTION

[0020] While specific embodiments are given in the drawings and the following description, they do not limit the disclosure. On the contrary, they provide the foundation for one of ordinary skill to discern the alternative forms, equivalents, and modifications that are encompassed in the scope of the appended claims.

[0021] For context, FIG. **1** shows an illustrative network such as might be found in a data processing center, with multiple server racks **102-106** each containing multiple servers **110** and at least one “top of rack” (TOR) switch **112**. The TOR switches **112** are connected to aggregator switches **114** for interconnectivity and connection to the regional network and internet. (As used herein, the term “switch” includes not just traditional network switches, but also routers, network bridges, hubs, and other devices that forward network communication packets between ports.) Each of the servers **110** is connected to the TOR switches **112** by network cables **120**, which may convey signals at high symbol rates.

[0022] FIG. **2** shows an illustrative switch **112** with an application-specific integrated circuit (ASIC) **202** that implements packet-switching functionality coupled to port connectors **204** for line cards or “pluggable modules” **206**. Pluggable modules **206** couple between the port connectors **204** and cable connectors **208** to improve communications performance by way of equalization and optional format conversion (e.g., converting between electrical and optical signals). The pluggable modules **206** may comply with any one of various pluggable module standards including SFP, SFP-DD, QSFP, QSFP-DD, and OSFP. Alternatively, the cables themselves may have connectors that conform to the pluggable module standards and incorporate the pluggable module circuitry.

[0023] The pluggable modules **206** may each include a retimer chip **210** and a microcontroller chip **212** that controls operation of the retimer chip **210** in accordance with firmware and parameters that may be stored in nonvolatile memory **214**. The operating mode and parameters of the pluggable retimer modules **206** may be set via a two-wire bus such as I2C or MDIO that connects the microcontroller chip **212** to the host device (e.g., switch **112**). The microcontroller chip **212** responds to queries and commands received via the two-wire bus, and responsively retrieves information from and saves information to control registers **218** of the retimer chip **210**.

[0024] Retimer chip **210** includes a host-side transceiver **220** coupled to a line-side transceiver **222** by first-in first-out (FIFO) buffers **224**. Though only a single lane is shown in the figure, the transceivers may support multiple lanes conveyed via multiple corresponding optical fibers or electrical conductors. A controller **226** coordinates the operation of the transceivers in accordance with the control register contents and may provide for multiple communication phases pursuant to a communications standard such as the Fibre Channel Standard published by the American National Standard for Information Technology Accredited Standards Committee INCITS, which provides phases for link speed negotiation (LSN), equalizer training, and normal operation.

[0025] The receiver portion of each transceiver may employ any of the many equalization and demodulation techniques disclosed in the open literature for recovering digital data from the degraded receive signal even in the presence of ISI. A critical piece of such techniques is a determination of the correct sample timing, as sample timing directly affects the signal to noise ratio of the discrete samples. Strategies for detecting and tracking optimal sample times exist with varying degrees of tradeoff between simplicity and performance, including those disclosed by the present inventors in U.S. Pat. No. 10,892,763, “Second-order clock recovery using three feedback paths”, which is hereby incorporated herein in its entirety.

[0026] FIGS. **3**A-**3**C show various clock recovery techniques that may be implemented by an illustrative receiver. Each of these receivers include an analog-to-digital converter **304** or other sampling element that samples the analog receive signal **302** at sample times corresponding to transitions in a sampling signal **305**, thereby providing a digital receive signal to a detection module **306**. The detection module **306** may apply equalization as well as symbol detection using, e.g., a matched filter, a decision feedback equalizer, a maximum likelihood sequence estimator, or any other suitable demodulation technique. The resulting stream of detected symbols **308** may be provided as a parallelized symbol stream for handling by “on-chip” circuitry, e.g., error correction and FIFO buffering.

[0027] The detection module **306** includes some form of a timing error estimator to generate an estimated timing error signal **310**. Any suitable design may be used for the timing error estimator including, e.g., a bang-bang or proportional phase detector. One suitable timing error estimator is set forth in co-owned U.S. Pat. No. 10,447,509, “Precompensator-based quantization for clock recovery”, which is hereby incorporated herein by reference in its entirety. Other suitable timing error estimators can be found in the open literature, including, e.g., Mueller, “Timing Recovery in Digital Synchronous Data Receivers”, IEEE Trans. Commun., v24n5, May 1976, and Musa, “High-speed Baud-Rate Clock Recovery”, U. Toronto Thesis, 2008.

[0028] In FIG. **3**A, the timing error signal **310** is coupled via a feedback path to control a phase interpolator **314** in a fashion that statistically minimizes the timing error signal **310**. In the feedback path, the timing error signal is scaled by a phase coefficient (K.sub.P) and integrated by a phase error accumulator **312** to obtain a phase code signal (supplied as a control signal to the phase interpolator **314**). The phase interpolator **314** operates on a clock signal from a phase lock loop (PLL) **316**. The phase interpolator **314** may receive or construct different phases of the clock signal, combining the different phases in accordance with the control signal to provide the sampling signal **305** having an interpolated phase more precisely matched to the symbols conveyed by the receive signal **302**.

[0029] The clock signal produced by phase lock loop (PLL) **316** is a frequency-multiplied version of a reference clock signal from reference oscillator **318**. A voltage-controlled oscillator (VCO) **320** supplies the clock signal to both the phase interpolator **314** and to a counter **322** that divides the frequency of the clock signal by a constant modulus N. The counter **322** supplies the divided-frequency clock signal to a phase-frequency detector (PFD) **324**. PFD **324** may use a charge pump (CP) as part of determining which input (i.e., the divided-frequency clock signal or the reference clock signal) has transitions earlier or more often than the other. A low pass filter **326** filters the output of PFD **324** to provide a control voltage to VCO **320**. The filter parameters are chosen so that the divided frequency clock becomes phase aligned with the reference oscillator.

[0030] The phase interpolator **314** is configured to interpolate the phase in discrete steps from 0 to 360°. A 7-bit control signal would enable **128** steps for a phase resolution of ˜2.8°. Other resolutions would also be suitable. In any case, the phase error accumulator **312** may be implemented with a higher bit resolution for improved performance, with the control signal being derived using truncation of the least significant bit(s). For various implementation reasons, the 360° interpolation range of the phase interpolator **314** may correspond to multiple symbol intervals, e.g., four, consequently reducing the number of phase increments per unit interval to, e.g., 32. The sub-unit phase may be determined by excluding the most significant bit(s) of the phase interpolator control signal.

[0031] For at least some contemplated uses, the reference oscillator **318** used by the receiver will often drift relative to the reference clock used by the transmitter and may differ by hundreds of ppm. To mitigate this issue, the receiver of FIG. **3**B includes a second feedback path in which the timing error signal is scaled by a frequency coefficient (K.sub.F) and integrated by a frequency error accumulator **330** to obtain a frequency offset signal. A summer **332** adds the frequency offset signal to the scaled timing error signal, supplying the sum to the phase error accumulator **312**. The control signal produced by the phase error accumulator **312** compensates for both the frequency offset and phase error of the clock signal relative to the analog receive signal **302**, thereby phase-aligning the sampling signal **305** with the data symbols in the analog receive signal **302**.

[0032] In both FIGS. **3**A and **3**B, any frequency offset between the PLL's clock signal output and the analog data signal is corrected by a continuous phase rotation of the phase interpolator **314**. This mode of operation imposes stringent demands on the linearity of the phase interpolator **314** over its entire tuning range, as the interpolator will repeatedly cycle through each of the phase interpolations during the continuous rotation. Any phase interpolation nonlinearity exhibits as periodic jitter in the sampling signal **305**, which becomes particularly exaggerated when the incoming signal employs spread spectrum clocking (SSC) to reduce electromagnetic interference.

[0033] The receiver in FIG. **3**C employs a third feedback path that may help address this issue. The receiver retains the analog-to-digital converter **304** for sampling the analog receive signal **302** and providing a digital receive signal to the detection module **306**. As before, the detection module **306** incudes a timing error estimator that generates a timing error signal **310**, and a first feedback path with the phase coefficient (K.sub.P) scaling and phase error accumulator **312**. In the second feedback path, the frequency error accumulator **330** (FIG. **3**B) is replaced with a modified frequency error accumulator **331** that is a leaky integrator that multiplies the accumulated frequency error by (1−K.sub.L) in each integration cycle. The leakage coefficient (K.sub.L) represents a gradual memory loss which, while it enables the second feedback path to provide a fast response, causes the frequency offset signal to tend toward zero over longer time scales.

[0034] The PLL **316** is replaced with a fractional-N phase lock loop **317** controlled via a third feedback path, enabling any frequency offset to be corrected separately from the phase interpolator **314**. The third feedback path includes a division-ratio scaling coefficient (K.sub.D) and a division-ratio error accumulator **334**, which supplies a division-ratio control signal to the fractional-N phase lock loop **317**. The fractional-N phase lock loop **317** is used in place of the original phase lock loop **316** to provide fine-grained frequency control of the clock signal supplied to the phase interpolator **314**. The division-ratio control signal adjusts the frequency offset of the clock signal relative to the data in the analog receive signal **302**, substantially reducing the phase rotation rate needed from the phase interpolator **314**.

[0035] A comparison of FIGS. **3**B & **3**C shows that the phase lock loop **316** and the fractional-N phase lock loop **317** both employ a PFD/CP **324** (comparing a divided frequency clock signal to the reference clock), low pass filter **326** (filtering the error to reduce noise), and a voltage-controlled oscillator **320** (supplying the output clock signal). Rather than dividing the output clock signal with a fixed modulus divider **322**, the fractional-N phase lock loop **317** uses a multi-modulus divider **323** that divides by N+m where N is a fixed large integer and m is a varying sign and small-magnitude integer (e.g. 0, +1, −1, +2, −2) given by the modulus selection signal at the end of (or, in alternative embodiments, at the beginning of, or at any point during) a count cycle. A delta-sigma modulator (DSM) **328** converts the division-ratio control signal into the modulus selection signal. The average value of N+m controls what fractional value the divider implements, enabling very fine control of the clock frequency supplied to the interpolator **314**.

[0036] The division-ratio error accumulator **334**, in combination with the low pass filter **326** of the phase lock loop **317**, operates on the longer time scale to overcome the memory loss of the modified accumulator **331**. Under steady-state or slow-changing conditions, the frequency offset correction is provided by the third feedback path, minimizing any effect of the phase interpolator nonlinearities. Where conditions where the frequency offset changes more quickly, the more transient corrections are provided by the first and second feedback paths.

[0037] Yet another illustrative implementation of the receiver omits the frequency error accumulator **331** of FIG. **3**C and relies solely on the division-error ratio accumulator **334** to correct any frequency offset. This additional implementation can be understood as setting the frequency coefficient K.sub.F to zero in FIG. **3**C.

[0038] Note that the frequency offset signals provided by accumulator **330** and the division ratio control signal provided by accumulator **334** may each qualify as a frequency signal as that term is used in the claims. Though expressed in different forms and determined relative to the reference oscillator frequency, they each represent a determination of the symbol frequency in the receive signal.

[0039] FIG. **4** shows an illustrative retimer that combines the receiver design of FIG. **3**C with a buffer **402** and transmitter **404**. Buffer **402** stores detected symbols **308** in a first-in first-out (FIFO) fashion. Transmitter **404** retrieves the buffered symbol stream **406** and retransmits it as an outgoing data stream **408**. A PLL **410** provides a transmit clock **411** to the transmitter **404** to control the symbol timing in the outgoing data stream **408**. The FIFO may accept sample clock **305** as a write clock for storing detected symbols **308** and may accept transmit clock **411** as a read clock for providing buffered symbols to the transmitter **404**. Clock recovery module **412** represents the filters and feedback path(s) circuitry that determines a frequency signal as part of converting the phase error signal **310** into a phase control signal for interpolator **314** and an optional division ratio control signal for PLL **317**.

[0040] The PLL **410** derives the transmit clock from the sample clock signal **305** and is adversely affected by jitter in the sample clock signal. It is desired to minimize jitter in the outgoing data stream without unduly increasing complexity, cost, or susceptibility to PVT variations.

[0041] FIG. **5** is a block diagram of an illustrative digital retimer that implements a retiming method with dual path clock forwarding filter **502** supplying a division ratio control signal to a transmit PLL **517**. Using the division ratio control signal, the transmit PLL **517** derives the transmit clock **411** from a reference clock supplied by reference oscillator **318**. Transmit PLL **517** may be implemented in a similar fashion as receive PLL **317** (FIG. **3**C).

[0042] Clock forwarding filter **502** includes a phase comparator **510** that compares the sampling clock **305** (supplied as a write clock to FIFO buffer **402**) with the transmit clock **411** (supplied as a read clock to FIFO buffer **402**) to derive a phase error signal **512**. The comparator **510** may use phase-frequency detector design or another suitable phase comparator technique. A phase filter **514** operates on the phase error signal to produce a filtered phase error signal that statistically minimizes variation of the phase error signal. Clock forwarding filter **502** is a dual path filter that also includes a frequency filter **516**. The frequency filter **516** operates on a frequency signal from the clock recovery module **412** to produce a filtered frequency signal. A summer sums the filtered phase error signal with the filtered frequency signal to produce the division ratio control signal. The frequency signal used by the frequency filter **516** in this case is the division ratio control signal for receiver PLL **317**.

[0043] Clock forwarding filter **502** may have the illustrative implementation shown in FIG. **6**. Phase comparator **510** includes a counter/frequency divider **520** for the reference clock (e.g., sampling clock **305**) and a second counter/frequency divider **521** for the local clock (e.g., transmit clock **411**), each operating to reduce the clock frequency by a given modulus N. An edge comparator **522** compares transitions of the divided clocks to determine whether the phase error is positive or negative, and optionally to also determine a magnitude of the phase error. Phase filter **514** includes a jitter filter **530**, shown here as an IIR filter that scales a current phase error by a jitter coefficient K.sub.J **531** and scales an accumulated value from register **532** by a coefficient **533** that equals 1−K.sub.J. The scaled values are summed and stored in the register **532** for the next clock cycle. The phase filter **514** further includes a frequency coefficient K.sub.RF **534** that scales the accumulated value from register **532** before summing it with a frequency offset from register **535**. The summed value is accumulated in register **535**. A phase coefficient K.sub.RP **536** scales the accumulated value from register **532** before summing it with the frequency offset from register **535**, producing a filtered phase error signal. The combination of the jitter filter **530** and the frequency offset accumulator (register **535**) make the phase filter **514** a second-order filter.

[0044] The frequency filter **516** is shown having a broadband gain coefficient K.sub.BB **540** and a bandwidth coefficient K.sub.B that scale the frequency signal (shown here as the receive PLL division ratio) before it is summed with an accumulated value from register **542** scaled by a coefficient (1−K.sub.B) **543**. The sum is stored as an accumulated value in register **542**. The accumulated value in register **542** is provided as a filtered frequency signal to a summer that adds the filtered frequency signal to the filtered phase signal, producing a frequency control signal (shown here as the transmit PLL division ratio).

[0045] FIGS. **5** and **6** merely show one illustrative implementation. As an example of an alternative implementation, the phase error signal **512** determined by phase comparator **510** can instead be determined as shown in FIG. **7**. FIG. **7** shows an illustrative implementation of FIFO buffer **402** having a counter **702** that converts the write clock into a write address pointer for a memory **704** having independent read and write ports. The memory stores the write data at the memory address indicated by the write address pointer. Another counter **706** converts the read clock into a read address pointer for the memory **704**, causing the data stored at the indicated read address to be provided for reading. A calculation circuit **708** subtracts the read address pointer from the write address pointer to determine the number of symbols stored in the memory **704**. (To address counter rollovers, a rollover flag may be set when the write counter rolls over and reset when the read counter rolls over. The subtraction element may treat the rollover flag as an extra bit for the write address pointer.) The calculation circuit may then subtract a target value from the number of stored symbols, providing the difference as the phase error signal **512**.

[0046] Other dual path clock forwarding filter implementations are also contemplated. FIG. **8** shows an illustrative retimer implementation in which the clock recovery module **412** determines a frequency offset rather than a PLL division ratio. (Referring momentarily to FIG. **3**B, this frequency offset may be provided by accumulator **330**.) The clock forwarding filter **802** includes the phase comparator **510**, phase filter **514**, and frequency filter **516** described previously in reference to FIG. **5**, though the frequency filter **516** operates on the frequency offset from clock recovery module **412**. A summer adds the filtered frequency signal from the frequency filter **516** to the filtered phase signal from phase filter **514** to obtain a phase offset signal **803**. A phase accumulator **804** may be provided as part of the clock forwarding filter **802** to derive a phase control signal for the transmit clock phase interpolator **814**. A transmit PLL **816** supplies a frequency-multiplied version of the reference clock signal from reference oscillator **318** to the transmit clock phase interpolator **814**. Based on the phase control signal, the transmit clock phase interpolator **814** derives the transmit clock **411** from the frequency-multiplied clock signal.

[0047] FIG. **9** shows another illustrative retimer implementation. As contrasted with the implementations of FIGS. **5** and **8**, the clock forwarding filter **902** supplies control signals to both the transmit PLL **917** and to the transmit clock phase interpolator **814**. The phase comparator **510**, phase filter **514**, and phase accumulator **804** derive a phase control signal for the transmit clock phase interpolator **814** from the phase relationship between the read and write clocks of FIFO **402**. The frequency filter **516** operates on the division ratio signal from clock recovery module **412** to obtain a filtered division ratio signal. The clock forwarding filter **902** provides the filtered division ratio signal to the transmit PLL **917**.

[0048] The illustrative retimer implementation of FIG. **10** is similar to that of FIG. **9**, but the transmit PLL **1017** has been modified to include the transmit clock phase interpolator **814** within the transmit PLL feedback loop. As with the fractional-N phase lock loop **317** (FIG. **3**C), the multi-modulus divider **323** frequency divides the transmit clock **411** by N+m where m is a varying sign and small-magnitude integer of the modulus selection signal at the end of (or, in alternative embodiments, at the beginning of, or at any point during) a count cycle. A delta-sigma modulator (DSM) **328** converts the division-ratio control signal into the modulus selection signal. The average of N+m controls what fractional value the divider implements, enabling very fine control of the transmit clock frequency. With the transmit clock phase interpolator **814** included, the transmit PLL feedback loop can mitigate the effects of any phase interpolation nonlinearities.

[0049] The foregoing retimer implementations are illustrated as single channel retimers. Multichannel systems can be implemented with multiple versions of the illustrated retimers arranged in parallel. While it is possible to operate the multiple retimers independently, such operation may permit undesirable levels of channel skew. To limit such channel skew, multichannel retimers may designate one channel as the primary channel and align the transmit clocks of all secondary channels with that of the primary channel.

[0050] FIG. **11** shows an illustrative multichannel retimer implementation that corresponds to the single channel implementation of FIG. **5**. The receive signal **302** and corresponding outgoing data stream **408** may be designated as the primary channel and handled as discussed previously with respect to FIG. **5**. Another receive signal **1100** and outgoing data stream **1108** may be designated as a secondary channel. A secondary clock forwarding filter **1102** provides a division ratio control signal to a secondary transmit PLL **1117**, which supplies a secondary transmit clock **1111** for the outgoing data stream **1108**. The secondary clock forwarding filter **1102** is distinguished from the primary clock forwarding filter **502** only in that the phase comparator **510** determines a phase error between the primary and secondary transmit clocks **411**, **1111**, rather than between the write and read clocks of the FIFO buffer **402**. This enables the multichannel retimer to statistically align the primary and secondary transmit clocks without forcing a rigid duplication.

[0051] FIG. **12** shows an illustrative multichannel retimer implementation that corresponds to the single channel implementation of FIG. **8**. The receive signal **302** and corresponding outgoing data stream **408** may be designated as the primary channel and handled as discussed previously with respect to FIG. **8**. Another receive signal **1100** and outgoing data stream **1108** may be designated as a secondary channel. A secondary clock forwarding filter **1202** provides a phase control signal to secondary transmit clock phase interpolator **1214**, which supplies a secondary transmit clock **1111** for the outgoing data stream **1108**. The secondary clock forwarding filter **1202** is distinguished from the primary clock forwarding filter **802** only in that the phase comparator **510** determines a phase error between the primary and secondary transmit clocks **411**, **1111**, rather than between the write and read clocks of the FIFO buffer **402**.

[0052] FIG. **13** shows an illustrative multichannel retimer implementation that corresponds to the single channel implementation of FIG. **9**. The receive signal **302** and corresponding outgoing data stream **408** may be designated as the primary channel and handled as discussed previously with respect to FIG. **9**. Another receive signal **1100** and outgoing data stream **1108** may be designated as a secondary channel. A secondary clock forwarding filter **1302** provides a division ratio control signal to secondary transmit PLL **1117** and a phase control signal to secondary transmit clock phase interpolator **1214**. Together transmit PLL **1117** and transmit clock phase interpolator **1214** produce a secondary transmit clock **1111** for the outgoing data stream **1108**. The secondary clock forwarding filter **1302** is distinguished from the primary clock forwarding filter **902** only in that the phase comparator **510** determines a phase error between the primary and secondary transmit clocks **411**, **1111**, rather than between the write and read clocks of the FIFO buffer **402**.
[0053] The foregoing integrated retimer circuits would typically be created using masks for patterning layers on semiconductor substrates during an integrated circuit manufacturing process. The mask patterns can be generated using commercially available software for converting the circuit schematics (usually expressed using a hardware description language such as Verilog) into semiconductor process masks. The circuits may be sub-units of more complex integrated circuit devices whose designs have been built up from modular components in a design database which resides on a nontransient information storage medium. Once the circuits are fully designed, software may convert the integrated circuits into semiconductor mask patterns also stored on a nontransient information storage medium and conveyed to the various process units in a suitable assembly line of an integrated circuit manufactory.
[0054] To facilitate the use of the disclosed retimers and retiming methods employing dual path clock forwarding filters, electronic device designers may express them as predefined modular units of integrated circuit layout designs. The designers can then arrange and join the modular units as needed to implement the various functions of the desired device. Each modular unit has a defined interface and behavior that has been verified by its creator. Though each modular unit may take a lot of time and investment to create, its availability for re-use and further development cuts product cycle times dramatically and enables better products. The predefined units can be organized hierarchically, with a given unit incorporating one or more lower-level units and in turn being incorporated within higher-level units. Many organizations have libraries of such predefined modular units for sale or license, including, e.g., embedded processors, memory, interfaces for different bus standards, power converters, frequency multipliers, sensor transducer interfaces, to name just a few. The predefined modular units are also known as cells, blocks, cores, and macros, terms which have different connotations and variations (“IP core”, “soft macro”) but are frequently employed interchangeably.
[0055] The modular units can be expressed in different ways, e.g., in the form of a hardware description language (HDL) file, or as a fully routed design that could be printed directly to a series of manufacturing process masks. Fully routed design files are typically process-specific, meaning that additional design effort would usually be needed to migrate the modular unit to a different process or manufacturer. Modular units in HDL form require subsequent synthesis, placement, and routing steps for implementation, but are process-independent, meaning that different manufacturers can apply their preferred automated synthesis, placement, and routing processes to implement the units using a wide range of manufacturing processes. By virtue of their higher-level representation, HDL units may be more amenable to modification and the use of variable design parameters, whereas fully routed units may offer better predictability in terms of areal requirements, reliability, and performance. While there is no fixed rule, digital module designs are more commonly specified in HDL form, while analog and mixed-signal units are more commonly specified as a lower-level, physical description.
[0056] Numerous alternative forms, equivalents, and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. For example, the integration-based

accumulators described herein can be replaced with other recursive or moving-average filter implementations providing a low-pass filter response. It is intended that the claims be interpreted to embrace all such alternative forms, equivalents, and modifications that are encompassed in the scope of the appended claims.

Claims

1. An integrated retimer circuit that comprises: a sampling element configured to produce a digital receive signal by sampling an analog receive signal in accordance with a sampling signal; a timing error estimator configured to produce a timing error signal indicating an estimated timing error of the sampling signal relative to the analog receive signal; a clock recovery circuit configured to derive the sampling signal from the estimated timing error and a reference clock in part by determining a frequency signal; a transmitter configured to retransmit the digital receive signal in accordance with a transmit clock; and a transmit clock generator configured to derive the transmit clock based on each of: the reference clock; the frequency signal; and a phase error of the transmit clock relative to the sampling signal.
2. The integrated retimer circuit of claim 1, wherein the transmit clock generator includes a clock forwarding filter having: a phase comparator to determine the phase error; a phase filter coupled to the phase comparator to derive a filtered phase error; and a frequency filter to derive a filtered frequency signal.
3. The integrated retimer circuit of claim 2, wherein the frequency signal is a division ratio for a fractional-N phase lock loop in a clock signal path for deriving the sampling signal from the reference clock, and wherein the transmit clock generator includes a fractional-N phase lock loop in a clock signal path for deriving the transmit clock from the reference clock.
4. The integrated retimer circuit of claim 2, wherein the frequency signal is a frequency offset for a phase interpolator in a clock signal path for deriving the sampling signal from the reference clock, and wherein the transmit clock generator includes a phase interpolator in a clock signal path for deriving the transmit clock from the reference clock.
5. The integrated retimer circuit of claim 4, wherein the clock signal path for deriving the sampling signal includes a fractional-N phase lock loop configured to receive a division ratio from the clock recovery circuit, and wherein the transmit clock generator includes division ratio filter configured to provide a filtered division ratio to a fractional-N phase lock loop in the clock signal path for deriving the transmit clock.
6. The integrated retimer circuit of claim 2, wherein the phase filter includes a jitter filter.
7. The integrated retimer circuit of claim 6, wherein the phase filter is a second-order filter.
8. The integrated retimer circuit of claim 2, wherein the phase error is based on a difference between a write pointer for a buffer and a read pointer for the buffer.
9. The integrated retimer circuit of claim 2, further comprising a secondary clock forwarding filter that aligns a secondary transmit clock to said transmit clock.
10. A retiming method that comprises: sampling an analog receive signal in accordance with a sampling signal to obtain a digital receive signal; producing a timing error signal indicating an estimated timing error of the sampling signal relative to the analog receive signal; deriving the sampling signal from the estimated timing error and a reference clock in part by determining a frequency signal; retransmitting the digital receive signal in accordance with a transmit clock; and deriving the transmit clock based on each of: the reference clock, the frequency signal, and a phase error of the transmit clock relative to the sampling signal.
11. The retiming method of claim 10, wherein a transmit clock generator derives the transmit clock using a clock forwarding filter having: a phase comparator to determine the phase error; a phase filter coupled to the phase comparator to derive a filtered phase error; and a frequency filter to derive a filtered frequency signal.

12. The retiming method of claim 11, wherein the frequency signal is a division ratio for a fractional-N phase lock loop in a clock signal path for deriving the sampling signal from the reference clock, and wherein the transmit clock generator includes a fractional-N phase lock loop in a clock signal path for deriving the transmit clock from the reference clock.
13. The retiming method of claim 11, wherein the frequency signal is a frequency offset for a phase interpolator in a clock signal path for deriving the sampling signal from the reference clock, and wherein the transmit clock generator includes a phase interpolator in a clock signal path for deriving the transmit clock from the reference clock.
14. The retiming method of claim 11, wherein the phase filter includes a jitter filter.
15. The retiming method of claim 14, wherein the phase filter is a second-order filter.
16. The retiming method of claim 11, wherein the phase error is based on a difference between a write address pointer for a buffer and a read address pointer for the buffer.
17. A nontransient information storage medium that comprises semiconductor manufacturing process mask patterns to make an integrated retimer circuit including: a sampling element configured to produce a digital receive signal by sampling an analog receive signal in accordance with a sampling signal; a timing error estimator configured to produce a timing error signal indicating an estimated timing error of the sampling signal relative to the analog receive signal; a clock recovery circuit configured to derive the sampling signal from the estimated timing error and a reference clock by determining a frequency signal; a transmitter configured to retransmit the digital receive signal in accordance with a transmit clock; and a transmit clock generator configured to derive the transmit clock based on each of: the reference clock; the frequency signal; and a phase error of the transmit clock relative to the sampling signal.
18. The nontransient information storage medium of claim 17, wherein the transmit clock generator includes a clock forwarding filter having: a phase comparator to determine the phase error; a phase filter coupled to the phase comparator to derive a filtered phase error; and a frequency filter to derive a filtered frequency signal.
19. The nontransient information storage medium of claim 18, wherein the frequency signal is a division ratio for a fractional-N phase lock loop in a clock signal path for deriving the sampling signal from the reference clock, and wherein the transmit clock generator includes a fractional-N phase lock loop in a clock signal path for deriving the transmit clock from the reference clock.
20. The nontransient information storage medium of claim 18, wherein the frequency signal is a frequency offset for a phase interpolator in a clock signal path for deriving the sampling signal from the reference clock, and wherein the transmit clock generator includes a phase interpolator in a clock signal path for deriving the transmit clock from the reference clock.
21. The nontransient information storage medium of claim 20, wherein the clock signal path for deriving the sampling signal includes a fractional-N phase lock loop configured to receive a division ratio from the clock recovery circuit, and wherein the transmit clock generator includes division ratio filter configured to provide a filtered division ratio to a fractional-N phase lock loop in the clock signal path for deriving the transmit clock.